

IPOH, Malaysia

WMO: 486250

Lat: 4.5663N

Lon: 101.0930E

Elev: 131

StdP: 14.63

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 11	(b) 72.0	(c) 73.1	(d) 66.2	(e) 97.0	(f) 80.7	(g) 67.9	(h) 103.0	(i) 81.1	(j) 12.3	(k) 81.8	(l) 11.0	(m) 82.0	(n) 4.1	(o) 40	(p) 0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 15.2	(c) 95.1	(d) 79.0	(e) 93.6	(f) 79.3	(g) 93.2	(h) 79.4	(i) 82.4	(j) 90.9	(k) 81.7	(l) 90.1	(m) 81.1	(n) 89.4	(o) 5.8	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.4	158.8	88.1	79.2	152.1	86.6	79.0	151.1	86.3	46.4	90.1	45.6	91.3	45.0	89.7	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.6	(b) 9.8	(c) 8.7	(e) 70.8	(f) 96.8	(g) 1.4	(h) 1.4	(i) 69.8	(j) 97.8	(k) 69.0	(l) 98.6	(m) 68.2	(n) 99.3	(o) 67.2	(p) 100.3		
(a) 11.6	(b) 9.8	(c) 8.7	(e) 67.3	(f) 83.9	(g) 1.8	(h) 0.9	(i) 66.0	(j) 84.5	(k) 64.9	(l) 85.0	(m) 63.9	(n) 85.5	(o) 62.6	(p) 86.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.1	82.3	83.4	83.9	83.8	83.9	83.8	83.4	83.2	82.7	82.5	82.2	81.9
	DBStd	1.80	1.93	1.69	1.90	1.68	1.40	1.62	1.79	1.63	1.64	1.37	1.41	1.70
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12075	1000	936	1052	1015	1051	1013	1036	1028	981	1007	966	990
	CDD65	6600	535	516	587	565	586	563	571	563	531	542	516	525
	CDH74	64021	4742	5129	5879	5600	6177	6138	5966	5706	5074	4858	4293	4459
	CDH80	25757	1844	2237	2545	2302	2524	2645	2489	2350	1957	1799	1498	1567
Wind	WSAvg	3.7	3.8	3.8	3.7	3.6	3.7	3.8	3.7	3.8	3.8	3.7	3.4	3.6
Precipitation	PrecAvg	94.0	5.9	5.2	6.5	9.8	8.4	5.7	6.4	6.1	8.2	11.7	10.8	9.3
	PrecMax	120.8	16.3	11.6	13.3	23.7	18.5	15.4	13.6	12.3	15.3	22.9	19.6	17.2
	PrecMin	75.3	0.6	0.8	0.8	3.8	3.0	1.1	1.5	2.0	2.1	3.6	2.8	3.2
	PrecStd	10.6	3.4	2.8	2.8	3.7	3.9	3.1	3.1	3.1	3.3	4.5	3.5	3.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.6	96.4	96.8	95.3	94.3	95.0	95.1	94.7	93.6	93.4	92.9	93.3
		MCWB	78.7	76.4	78.8	80.4	80.7	79.0	78.9	78.7	78.4	79.3	79.9	80.0
	2%	DB	91.8	93.6	94.9	93.4	93.1	93.3	93.2	93.0	91.8	91.6	91.4	91.5
		MCWB	78.5	78.2	79.0	81.0	80.9	79.4	79.1	78.4	78.5	79.2	79.7	79.0
	5%	DB	91.3	93.0	93.2	91.7	91.6	91.7	91.6	91.5	91.1	90.0	89.8	89.9
		MCWB	78.5	78.4	79.4	80.6	80.6	79.5	79.0	78.4	78.4	78.9	79.5	78.7
	10%	DB	89.5	91.3	91.4	91.0	91.0	91.1	90.0	89.9	89.5	89.3	88.1	88.1
		MCWB	78.0	78.3	79.3	80.4	80.4	79.4	78.8	78.3	78.4	78.8	79.5	78.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.2	81.9	82.2	83.1	83.4	82.2	82.4	81.9	81.0	82.2	82.3	81.6
		MCDB	90.1	90.6	92.1	91.9	91.4	90.6	91.0	89.3	88.7	90.1	89.8	90.3
	2%	WB	80.1	80.8	81.1	81.9	82.1	81.3	81.0	80.2	80.1	80.7	81.0	80.3
		MCDB	88.7	89.9	90.6	90.6	90.1	89.4	89.1	88.3	87.9	88.7	88.7	88.4
	5%	WB	79.4	79.9	80.4	81.2	81.4	80.6	80.1	79.6	79.5	79.8	80.0	79.5
		MCDB	87.8	88.7	89.6	89.6	89.2	88.6	88.0	87.7	87.3	87.4	87.4	87.3
	10%	WB	78.7	79.2	79.7	80.4	80.6	79.9	79.4	79.0	78.9	79.1	79.2	78.8
		MCDB	86.8	87.5	88.3	88.4	88.2	87.6	87.3	87.1	86.7	86.3	86.2	86.2
Mean Daily Temperature Range	5% DB	MDBR	15.6	17.3	16.7	15.8	14.9	15.2	15.4	15.5	15.0	14.9	14.7	14.5
		MCDBR	17.7	19.0	18.3	16.8	15.9	16.3	16.5	16.8	16.4	16.1	16.1	16.6
		MCWBR	7.4	7.6	7.3	7.0	6.8	6.9	6.6	6.5	6.5	6.5	6.8	6.9
	5% WB	MCDBR	16.3	17.0	16.8	16.0	15.0	15.3	15.4	15.6	15.0	15.1	15.2	15.2
		MCWBR	7.2	7.3	7.2	7.0	6.8	6.9	6.9	6.6	6.5	6.6	6.7	6.7
Clear-Sky Solar Irradiance	taub	0.457	0.501	0.506	0.511	0.499	0.526	0.516	0.529	0.529	0.540	0.467	0.444	
	taud	2.299	2.156	2.161	2.154	2.199	2.124	2.148	2.100	2.103	2.069	2.290	2.358	
	Ebn at Noon	275	265	263	257	254	243	247	249	253	251	270	277	
	Edh at Noon	43	50	50	49	46	49	48	52	52	54	43	40	
All-Sky Solar Radiation	RadAvg	1479	1725	1758	1753	1642	1625	1594	1619	1565	1503	1423	1354	
	RadStd	115	118	133	53	58	69	69	59	87	110	65	105	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.70	+0.40	N/A	+0.94	+0.79	+0.67	N/A	N/A	+252	+252

Nomenclature: See separate page